

Haesung Kim

Ph.D. in Electrical Engineering

Postdoctoral Researcher

Specere Lab, School of Mechanical Engineering, Birk Nanotechnology Center, Purdue University, West Lafayette, IN

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Research Focus

Opto-electronics-based characterization of semiconductor devices, physics-based device modeling, and the development of parameter-extraction methods. The work centers on extracting device parameters that are otherwise difficult to separate — parasitic resistances, interface and subgap trap distributions, and the conduction band minimum — using opto-electronic and transient measurements, applied across silicon MOSFETs (including gate-all-around and NAND flash structures), amorphous IGZO thin-film transistors, ferroelectric FETs, and ferroelectric TFTs.

Education

2022.03 – 2025.08	Ph.D. in Electrical Engineering Kookmin University, Seoul, Korea Advisers: Prof. Dong Myong Kim, Prof. Jong-Ho Bae
2020.03 – 2022.02	M.S. in Electrical Engineering Kookmin University, Seoul, Korea
2014.03 – 2020.02	B.S. in Electrical Engineering Kookmin University, Seoul, Korea

Appointments

2025.08 – present	Postdoctoral Researcher Specere Lab, School of Mechanical Engineering Birk Nanotechnology Center, Purdue University, West Lafayette, IN, USA Advisor: Prof. Thomas Beechem
2025.03 – 2025.06	Lecturer Kookmin University, Seoul, Korea Intelligent Semiconductor Devices (3 credits, junior level)
2015.09 – 2017.06	Republic of Korea Marine Corps Mandatory military service

Journal Articles

1. Y. Choi, H. Yang, H. J. Kim, S. Park, Y. J. Lee, S. J. Choi, D. H. Kim, D. M. Kim, **H. Kim**, J. H. Bae, “Characterization of trap dynamics via transient response in amorphous InGaZnO thin-film transistors,” *Applied Physics Letters*, vol. 128, no. 24, 2026. doi:10.1063/5.0332203
2. H. J. Kim, H. Yang, **H. Kim**, S. Park, H. N. Lee, S. J. Choi, D. M. Kim, D. H. Kim, M. K. Park, J. H. Bae, “Analysis of spatial distribution of remnant polarization in inverted staggered a-IGZO/HZO ferroelectric thin-film transistor using capacitance-voltage characteristics,” *Solid-State Electronics*, vol. 235, pp. 109358, 2026. doi:10.1016/j.sse.2026.109358
3. **H. Kim**, J. Y. Park, S. H. Han, H. Yang, Y. J. Lee, S. J. Choi, D. H. Kim, J. H. Bae, D. M. Kim, “Characterization of Low-Temperature Intrinsic Field-Effect Mobility With Contact Resistances in Amorphous InGaZnO Thin Film Transistors,” *IEEE Transactions on Electron Devices*, vol. 73, no. 7, pp. 4341–4345, 2026. doi:10.1109/TED.2026.3691120
4. **H. Kim**, J. Park, J. Im, H. Kim, Y. M. Kim, K. Park, S. J. Choi, D. H. Kim, D. M. Kim, Y. J. Yoon, S. Y. Woo, J. H. Bae, “Impact of proton irradiation on SiNx and Si–SiO2 interfaces in FLASH memory cell,” *Nuclear Engineering and Technology*, vol. 58, no. 10, pp. 104433, 2026. doi:10.1016/j.net.2026.104433

5. Y. Kim, Y. J. Lee, J. Yang, B. Kim, S. J. Kim, J. Kim, I. Im, J. Y. Kim, H. Rui, **H. Kim**, C. W. Bark, M. H. Park, D. H. Kim, S. J. Choi, J. J. Yang, S. Lee, H. W. Jang, "Programmable ferroelectric rectifier for reliable and efficient neuromorphic crossbar array," *Nature Communications*, vol. 17, no. 1, 2026. doi:10.1038/s41467-026-70727-2
6. S. Kim, H. Yang, H. Lee, J. Park, H. Jeong, S. Y. Woo, M. K. Park, **H. Kim**, J. H. Bae, "Source/drain metal work-function-dependent memory characteristics governed by ferroelectric switching-assisted current modulation in a-IGZO FeFETs," *Materials Science in Semiconductor Processing*, vol. 215, pp. 111021, 2026. doi:10.1016/j.mssp.2026.111021
7. **H. Kim**, H. Yang, Y. Choi, H. Kim, Y. J. Lee, S. J. Choi, D. H. Kim, D. M. Kim, D. Kwon, J. H. Bae, "Unveiling charge trapping dynamics in SiO₂/HfZrOX-based ferroelectric field-effect-transistors to minimize read-after-write latency," *Current Applied Physics*, vol. 89, pp. 181–185, 2026. doi:10.1016/j.cap.2026.06.007
8. H. Lee, S. Kim, C. Han, **H. Kim**, H. Yang, S. Park, S. Yun, Y. J. Lee, S. J. Choi, D. H. Kim, D. M. Kim, D. Kwon, J. H. Bae, "Quantitative Analysis on the Interaction Between Channel Carrier and Remote Trap in Hf_xZr_{1-x}O₂/SiO₂ Interface in Ferroelectric Field-Effect-Transistor," *Advanced Electronic Materials*, vol. 12, no. 2, 2026. doi:10.1002/aelm.202500549
9. J. Park, **H. Kim**, S. J. Choi, D. H. Kim, D. M. Kim, J. H. Bae, "Exploring low-frequency noise dynamics in a-IGZO TFTs: Unveiling the impact of contact metal variations for advanced semiconductor applications," *Solid-State Electronics*, vol. 231, pp. 109271, 2026. doi:10.1016/j.sse.2025.109271 [†]
10. H. Yang, **H. Kim**, H. J. Kim, D. M. Kim, S. J. Choi, D. H. Kim, Y. J. Lee, J. H. Bae, "Defect-mediated enhancement of memory window in IGZO-channel ferroelectric field effect transistors," *Materials Science in Semiconductor Processing*, vol. 204, pp. 110268, 2026. doi:10.1016/j.mssp.2025.110268
11. **H. Kim**, S. H. Han, H. Jeong, H. Yang, S. J. Choi, D. H. Kim, J. H. Bae, D. M. Kim, "Photonic I–V Technique With Photogating Effect for Extended Extraction of Subgap Density of States in Amorphous Oxide Semiconductor TFTs," *IEEE Transactions on Electron Devices*, vol. 72, no. 5, pp. 2751–2755, 2025. doi:10.1109/TED.2025.3547707
12. S. Park, H. Yang, **H. Kim**, H. Jeong, S. J. Choi, D. H. Kim, D. M. Kim, M. K. Park, J. H. Bae, "Analysis of operation delay and switching speed limitation due to source/drain resistance and subgap density of states in amorphous InGaZnOx/HfZrOx ferroelectric thin-film transistor," *Solid-State Electronics*, vol. 229, pp. 109203, 2025. doi:10.1016/j.sse.2025.109203
13. S. H. Han, **H. Kim**, J. H. Bae, S. J. Choi, D. H. Kim, D. M. Kim, "Photovoltaic Effect De-Embedded Photonic C–V Characterization of Subgap Density of States in Amorphous Oxide Semiconductor Thin-Film Transistors," *IEEE Transactions on Electron Devices*, vol. 71, no. 11, pp. 6795–6798, 2024. doi:10.1109/TED.2024.3469161 [†]
14. **H. Kim**, H. Yang, S. Lee, S. Yun, J. Park, S. Park, H. N. Lee, H. Kim, S. J. Choi, D. H. Kim, D. M. Kim, D. Kwon, J. H. Bae, "Comprehensive analysis of read-after-write latency in HfZrOX-based ferroelectric field-effect-transistors with SiO₂ interfacial layer," *Applied Physics Letters*, vol. 124, no. 3, 2024. doi:10.1063/5.0172204
15. S. Lee, **H. Kim**, H. Yang, S. Yun, J. Park, H. Lee, S. Park, S. J. Choi, D. H. Kim, D. M. Kim, D. Kwon, J. H. Bae, "Analysis of the Role of Interfacial Layer in Ferroelectric FET Failure as a Memory Cell," *IEEE Electron Device Letters*, vol. 45, no. 4, pp. 562–565, 2024. doi:10.1109/LED.2024.3360419 [†]
16. H. Yang, S. Park, S. Yun, **H. Kim**, H. Lee, M. K. Park, S. J. Choi, D. H. Kim, D. M. Kim, D. Kwon, J. H. Bae, "Exploration of structural influences on the ferroelectric switching characteristics of ferroelectric thin-film transistors," *Nanoscale*, vol. 16, no. 42, pp. 19856–19864, 2024. doi:10.1039/d4nr02096k
17. **H. Kim**, H. B. Yoo, H. Lee, J. H. Ryu, J. Y. Park, S. H. Han, H. Yang, J. H. Bae, S. J. Choi, D. H. Kim, D. M. Kim, "Extraction Technique for the Conduction Band Minimum Energy in Amorphous Indium–Gallium–Zinc–Oxide Thin Film Transistors," *IEEE Transactions on Electron Devices*, vol. 70, no. 6, pp. 3126–3130, 2023. doi:10.1109/TED.2023.3269735 [†]
18. J. Y. Park, **H. Kim**, H. B. Yoo, J. H. Ryu, S. H. Han, J. H. Bae, S. J. Choi, D. H. Kim, D. M. Kim, "Simultaneous Extraction of Mobility Enhancement Factor and Threshold Voltage With Parasitic Resistances in Amorphous Oxide Semiconductor Thin-Film Transistors," *IEEE Transactions on Electron Devices*, vol. 70, no. 5, pp. 2312–2316, 2023. doi:10.1109/TED.2023.3253468 [†]
19. **H. Kim**, H. B. Yoo, J. H. Ryu, J. H. Bae, S. J. Choi, D. H. Kim, D. M. Kim, "Current-to-transconductance ratio technique for simultaneous extraction of threshold voltage and parasitic resistances in MOSFETs," *Solid-State Electronics*, vol. 183, pp. 108133, 2021. doi:10.1016/j.sse.2021.108133 [†]

20. H. B. Yoo, **H. Kim**, J. H. Ryu, J. Park, J. H. Bae, S. J. Choi, D. H. Kim, D. M. Kim, “Modeling and characterization of photovoltaic and photoconductive effects in insulated-gate field effect transistors under optical excitation,” *Solid-State Electronics*, vol. 186, pp. 108139, 2021. doi:10.1016/j.sse.2021.108139
21. J. Yu, H. B. Yoo, **H. S. Kim**, J. H. Ryu, S. J. Choi, D. H. Kim, D. M. Kim, “Characterization of Spatial Distribution of Trap Across the Substrate in Metal-Insulator-Semiconductor Structure with Band Bending Effect,” *Journal of Nanoscience and Nanotechnology*, vol. 21, no. 8, pp. 4315–4319, 2021. doi:10.1166/jnn.2021.19391
22. **H. Kim**, H. B. Yoo, J. T. Yu, J. H. Ryu, S. J. Choi, D. H. Kim, D. M. Kim, “Alternating Current-Based Technique for Separate Extraction of Parasitic Resistances in MISFETs With or Without the Body Contact,” *IEEE Electron Device Letters*, vol. 41, no. 10, pp. 1528–1531, 2020. doi:10.1109/LED.2020.3020405 [†]
23. H. B. Yoo, J. Yu, **H. Kim**, J. H. Ryu, S. J. Choi, D. H. Kim, D. M. Kim, “Deep depletion capacitance–voltage technique for spatial distribution of traps across the substrate in MOS structures,” *Solid-State Electronics*, vol. 173, pp. 107905, 2020. doi:10.1016/j.sse.2020.107905

Conference Papers (indexed)

1. H. Yang, **H. Kim**, C. Han, Y. J. Lee, S. J. Choi, D. H. Kim, D. M. Kim, D. Kwon, J. H. Bae, “Quantitative Analysis of Endurance-Dependent Charge Trapping Dynamics in Ferroelectric Field-Effect Transistors with Temperature Effects,” *2025 Silicon Nanoelectronics Workshop (SNW)*, pp. 74–75, 2025. doi:10.23919/SNW65111.2025.11097276
2. H. B. Yoo, **H. Kim**, Y. Lee, J. H. Ryu, J. Y. Park, H. J. Yang, J. H. Bae, D. H. Kim, S. J. Choi, D. M. Kim, “Characterization Technique for Interface Traps in Si Nanosheet GAA MOSFETs through Subthreshold I-V Characteristics,” *2022 IEEE 22nd International Conference on Nanotechnology (NANO)*, pp. 116–119, 2022. doi:10.1109/NANO54668.2022.9928778 [†]

Preprints

1. D. M. Kim, J. H. Ryu, H. B. Yoo, **H. Kim**, J. Y. Park, S. H. Han, J. H. Bae, S. J. Choi, D. H. Kim, “Capacitance-Based Extraction Technique for the Spatial and Energy Distributions of Traps in Si Mosfets,” *SSRN*, 2023. doi:10.2139/ssrn.4377528

[†] Co-first author.

Patents

2026 **Method and apparatus for extracting the conduction band minimum of amorphous oxide semiconductor thin-film transistors** (translated; official title in Korean)
 KR Patent **10-2914693** · registered 2026-01-14 · filed 2023-03-17 (10-2023-0035106)
 Assignee: Kookmin University Industry-Academic Cooperation Foundation · 11 claims · inventor 5 of 10

Research Grants and Projects

2024-09 – 2025-08	Principal Investigator — National Research Foundation of Korea Maximizing Energy Efficiency and Speed: Modeling and Optimization of Transient Response Characteristics in AOS-based Ferroelectric Memory Devices
2024-03 – 2024-12	Graduate Researcher — National Research Foundation of Korea Development of Application-customized Computing Be-spoke Device for High-density and Efficient PIM
2022-06 – 2023-05	Graduate Researcher — Industry–University Cooperation Research, Alsemy Co. Semiconductor device modeling using artificial neural networks
2020 – 2024	Graduate Researcher and Staff — National Research Foundation of Korea Four-Step Brain Korea 21 Project
2020 – 2022	Graduate Researcher — National Research Foundation of Korea Hybrid Device-Based Circadian ICT Research Center

Awards and Honors

2025	Excellent Talent Award , Kookmin University (Ph.D. conferral, Fall 2025)
2023	Excellent Poster Award on Site , The 30th Korean Conference on Semiconductors
2022	Excellent Talent Award , Kookmin University (M.S. conferral, Spring 2022)

Teaching Experience

2025	Lecturer — Intelligent Semiconductor Devices, Kookmin University (3 credits, junior level; chalkboard lecture. Reference: Neamen, Semiconductor Physics and Devices)
2024	Session Instructor — Winter job training: Measurement of semiconductor devices and extraction of characteristic parameters, Kookmin University (senior level)
2023	Session Instructor — Summer job training: Measurement of semiconductor devices and extraction of characteristic parameters, Kookmin University (senior level)
2023	Teaching Assistant — Basic Electronic Experiment, Kookmin University (sophomore level)
2023	Teaching Assistant — Electrical Engineering Capstone Design I, Kookmin University (senior level)
2022	Teaching Assistant — Electrical Engineering Capstone Design II, Kookmin University (senior level)
2022	Teaching Assistant — Electrical Engineering Capstone Design I, Kookmin University (senior level)
2021	Teaching Assistant — Electrical Engineering Capstone Design II, Kookmin University (senior level)
2021	Session Instructor — Summer job training: Semiconductor Device TCAD Simulation Training, Kookmin University (junior–senior level)
2020	Teaching Assistant — Electrical Engineering Capstone Design II, Kookmin University (senior level)
2019	Undergraduate Mentoring — Semiconductor Engineering, Kookmin University (junior level. Reference: D. M. Kim, Microelectronic Semiconductor Physics and Devices)
2019	Undergraduate Mentoring — Electromagnetics, Kookmin University (sophomore level. Reference: W. Hayt, Engineering Electromagnetics)

Selected Conference Presentations

- H. Kim, S. H. Han, H. Jeong, H. Yang, S.-J. Choi, D. H. Kim, J.-H. Bae, D. M. Kim, “Photonic I–V-based technique for subgap density of states extraction in AOS TFTs,” *International Conference on Electronics, Information, and Communication (ICEIC)*, Osaka, Japan, Jan. 2025.
- H. Kim, J. Park, J. Im, H. Kim, Y. J. Yoon, Y.-M. Kim, K. Park, S. Y. Woo, J.-H. Bae, “Impact of Proton Irradiation on SiNx and Si–SiO₂ Interfaces in FLASH Memory,” *IEEE Silicon Nanoelectronics Workshop (SNW)*, Honolulu, HI, USA 2024, pp. 1–2.
- J. Y. Park, H. Kim, H. B. Yoo, J. H. Ryu, J.-H. Bae, S.-J. Choi, D. H. Kim, D. M. Kim, “Comprehensive DC Technique for Extraction of Threshold Voltage, Parasitic Resistances, and Mobility Enhancement Parameter in AOS TFTs,” *The 28th Korean Conference on Semiconductors*, Jan. 2021. [†]
- H. B. Yoo, H. Kim, J. Yu, Y. J. Park, S.-J. Choi, D. H. Kim, D. M. Kim, “Alternating Current-based Open Drain Method for Separate Extraction of Source and Drain Resistances in MOSFETs,” *The 27th Korean Conference on Semiconductors*, Feb. 2020. [†] (presenter)

Conference Participation

International. International Conference on Electronics, Information, and Communication (Osaka, Japan), 2025; IEEE Silicon Nanoelectronics Workshop (Honolulu, HI, USA), 2024; IEEE International Conference on Nanotechnology (22nd) (Palma de Mallorca, Spain), 2022.

Domestic (Korea). The 31st Korean Conference on Semiconductors, 2024; The 30th Korean Conference on Semiconductors, 2023; Summer Annual Conference of the Institute of Electronics and Information Engineers, 2022; The 29th Korean Conference on Semiconductors, 2022; The 28th Korean Conference on Semiconductors, 2021; The 27th Korean Conference on Semiconductors, 2020 [oral].

Technical Skills

Measurement. Agilent 4156C / B1500A and Keithley 4200-SCS semiconductor parameter analyzers; HP 4284A and 4294A LCR meters; Agilent E5250A switching matrix; SUMMIT-9200 probe station; MS TECH MST-1000B hot chuck controller; laser diode sources (OZ-Series) and PM100D optical power meter; RIGOL DG5102 function generator; DPO3014 oscilloscope; Keysight 35670A dynamic signal analyzer; SR570 low-noise current preamplifier; CX3322A device current waveform analyzer; WITec alpha Raman/PL spectroscopy.

Simulation and computation. Synopsys ISE TCAD, Silvaco TCAD, OrCAD PSpice Designer, PSIM, P-CAD, MATLAB, Python.